The Camouflage Ratio: Strategic Tie Formation and the Limits of Structural Detection

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Abstract

Inferring an actor's attributes from those of its network neighbours is among the oldest uses of relational data, and it rests on homophily: neighbours are assumed to be informative. We ask what happens when an actor *chooses* its ties in order to defeat that inference. Automated accounts on social platforms do exactly this, forming ties to genuine users so that their local neighbourhood resembles a genuine one—a strategy known as relation camouflage. We formalise it with a single structural parameter, the *camouflage ratio* ρ, the share of an actor's ties directed at the other class, and ask when neighbourhood information still improves on attributes alone.

Three results follow. First, camouflage is not a private act: because a tie has two ends, an actor that camouflages necessarily contaminates its neighbours' neighbourhoods in turn. Edge-endpoint conservation forces the induced contamination of the majority to be $\eta = c\rho$, where c is the minority-to-majority ratio, so that *prevalence acts as protection*—the more common the strategic actors, the less camouflage each one needs to defeat relational inference. Second, this yields a threshold $\rho^\star = (1 - D^{-1/2})/(1 + c)$ in the mean neighbourhood size D, above which averaging over neighbours is worse than ignoring the network entirely, and a bounded interval above which relational information becomes informative again with its sign reversed: the most damaging strategy is *intermediate*, not maximal, camouflage. We are explicit that a crossover of this kind is not new—at $c = 1$ it recovers a result of Ma et al. (2022) and the interval is the second root of the same condition—and that our contribution is the asymmetric, class-imbalanced case together with a variance correction that shifts the threshold by 11.5%. Third, selectively discarding suspicious ties does not straightforwardly restore inference: it trades bias against the number of ties left to average over, and we bound the discrimination achievable from attribute discrepancies alone.

We then measure ρ on MGTAB, a Twitter dataset of 10,199 expert-annotated accounts in which every account is labelled, so that no tie is lost to unlabelled alters. Automated accounts turn out to be almost perfectly camouflaged on every social relation—the median one has *no* automated alter, and automated–automated ties number 212 against 40,263 automated–genuine ties—which places those relations at or beyond the *upper* edge of the harmful interval rather than inside it: a neighbourhood that is entirely genuine, in a population that is 27% automated, is itself anomalous. The relations that do defeat inference are content-coordination ties (shared URLs and hashtags) at intermediate $\rho \approx 0.6$. The same accounts therefore avoid one another where ties are visible and converge where content is shared, and ρ differs between the two layers by a factor approaching two on identical actors—so multiplexity, not aggregate density, decides whether relational inference is usable. We also document why the canonical bot benchmarks cannot support this measurement at all, and confirm the edge-endpoint identity to machine precision in its degree-corrected form.

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1. Introduction

One of the oldest uses of relational data is to infer something about an actor from the actors it is connected to. The inference rests on *homophily*: if ties are formed preferentially between similar actors, then a neighbourhood is informative about its centre, and averaging over it estimates the centre's attributes more precisely than the centre's own noisy attributes do. Variants of this logic underpin network autocorrelation models, peer-influence estimation, and relational classification.

This paper asks what happens when an actor *chooses* its ties in order to defeat that inference. The setting where this is most consequential, and where labelled data exists, is the detection of automated accounts on social platforms. Early generations of such accounts formed dense, mutually connected clusters and were consequently easy to identify collectively. Contemporary ones do the opposite: they form ties to large numbers of genuine users so that their local neighbourhood is statistically indistinguishable from a genuine actor's—a strategy known in that literature as *relation camouflage* [11, 12]. The strategy is purposive and cheap, and it targets the inferential mechanism rather than any particular detector. Detection systems have responded by discarding suspicious ties, by treating similar and dissimilar neighbours through separate channels, or by holding an actor's own attributes apart from its neighbours' [11, 13]; our own earlier work combines all three [1]. Those designs are justified empirically, by ablation. What has been missing is an account of the structural conditions under which the underlying inference works at all, which is what we supply here.

We formalise it with one structural parameter, the *camouflage ratio* ρ: the share of a strategic actor's ties directed at the other class. Two questions then have precise answers.

1. **When is relational information worth using at all?** If camouflage is severe enough, averaging over neighbours must eventually be worse than ignoring the network and using attributes alone. Practitioners have no criterion for where that point lies, and—as we show—it depends on quantities they can already measure.

2. **Does selectively discarding suspicious ties restore the inference?** Discarding ties is the standard remedy, and it is justified almost entirely by ablation: removing the step costs accuracy on a benchmark. That is not a statement about the conditions under which it works.

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Our analysis is conducted in a contextual stochastic block model [3]—a stochastic block model with attributes on the nodes—augmented with ρ as an explicit strategic parameter, and we study what neighbourhood averaging does to the separation between the two classes as ρ grows.

What is already known. We are explicit about priority from the outset. For a *balanced* symmetric CSBM, Ma et al. [5, Thm. 2] already established a closed-form threshold at which aggregation ceases to beat the graph-free classifier, and our critical ratio reduces to their condition exactly when the two classes are equal in size (Remark 9); related crossover analyses appear in [14–16]. The existence of a crossover is therefore not our discovery, and we do not claim it. Our contribution is the *class-imbalanced adversarial* case, where the two classes deflate by different amounts—which moves the decision boundary and makes bot prevalence itself a robustness parameter—together with a closed form for how far edge pruning shifts the threshold.

Contributions.

- **The edge-balance identity, and bot prevalence as a robustness parameter.** Camouflage edges have two ends, so a bot that points an edge at a human necessarily pollutes that human's neighbourhood too. We show this forces $\eta = c\rho$, where c is the bot-to-human ratio, so that the total deflation is $1 - (1 + c)\rho$ (Proposition 4). The consequence is that bot prevalence enters the robustness threshold directly: as bots become more common, each one needs *less* camouflage to break the detector. This is not a class-imbalance problem that resampling addresses.

- **A Bayes-error crossover valid under class imbalance.** When $c \neq 1$ the two classes deflate asymmetrically, the post-aggregation class midpoint moves, and a comparison against a fixed decision boundary is no longer valid—the objection Luan et al. [15] raise against [5]. We compare Bayes errors instead (Theorem 5), obtaining $\rho^\star = (1 - D^{-1/2})/(1 + c)$ (Corollary 8), which recovers [5] at $c = 1$ and [2] at $\rho = 0$. Our error term also retains a mixture-variance contribution that a naive σ^2/D calculation misses, and which shifts the threshold by 11.5%.

- **What discarding ties can and cannot buy.** Selectively removing suspicious ties is the standard remedy, and we show it is a trade rather than a free improvement: down-weighting ties non-uniformly also reduces the number of ties left to average over, so the benefit is bounded (Proposition 14). We give the tolerance as a closed function of the discriminating power of whatever suspicion score is used, and bound that power from the distribution of attribute discrepancies, which removes the free parameter (Section 4.4). On the review networks we measure, a score based on attribute discrepancy alone moves the threshold by less than 0.01 and does not restore the inference, while a flawless one would—so the operative quantity is the gap between achievable and ideal tie scoring.

- **Measurement on real networks.** We give an estimator for ρ, validate it against planted ground truth, and report it for several labelled populations. Camouflage differs by a factor approaching two across relation types recorded on the *same* actors, which makes these per-relation quantities and implies that pooling relations averages a usable signal with an unusable one.

Our aim is not to propose a detector. It is to characterise a structural mechanism that limits what ties can reveal, to show that its governing quantities are measurable, and to report what they are on real networks. The practical consequence is that "should I use the network here?" becomes a question that can be answered from data before any model is fitted, and answered separately for each type of tie.

2. Related Work

2.1. Inferring attributes from alters

The operation we analyse has a long history in network analysis under several names. In the network autocorrelation model an actor's outcome is regressed on a weighted average of its alters' outcomes, and Leenders [33] makes the point on which this paper turns: the weight matrix is a substantive modelling commitment, not a technical detail, and the conclusions depend on it. Row-normalised weights—the natural choice, and the one implicit in most applications—are exactly the mean aggregation of Section 3. Everett and Borgatti [32] give the general matrix formulation of *alter composition*, the profile of categories an actor is exposed to through its alters, and extend it to overlapping memberships; our camouflage ratio is a two-category alter-composition measure, and the question we ask is what happens to it when one category chooses its ties.

Two in-journal results bear directly on the reliability of this inference. Roeling and Nicholls [34] impute unobserved attributes from network structure and find the procedure sensitive to model misspecification and to missingness. Vörös et al. [35] report a negative result of the same kind: across five countries, friendship relations turn out to be a poor proxy for status attributions, so one relation cannot be substituted for another. Both are warnings that neighbour-based inference is conditional rather than automatic; our contribution is to give one condition in closed form, for the case where the failure is *induced deliberately*.

Homophily is the assumption the inference rests on, and its measurement is a developed literature here: Bojanowski and Corten [38] survey and axiomatise segregation and homophily indices, and McMillan [39] shows that homophily measured on strong and weak ties differs in magnitude, so that a single homophily figure for a population can be misleading. Noel and Nyhan [37] demonstrate the consequence for inference, showing that homophily in tie *retention* biases estimates of social influence. Our results are the adversarial limit of this concern: we treat the degree of homophily violation as a decision variable of one class rather than a property of the population.

2.2. Multiplexity, and why the analysis is per relation

That different relations among the same actors have different structure is a founding observation of this journal [41], and its methodological consequence—that collapsing relations into a single adjacency destroys information, while modelling them jointly quickly becomes intractable—is treated directly by Vörös and Snijders [42]. Snijders et al. [43] show the stronger version: structure apparently present in an advice relation is substantially mediated by the friendship relation beneath it. An et al. [36] find the same phenomenon quantitatively, with peer effects differing in magnitude between friends and classmates in a multilayer setting.

This matters for us empirically rather than only in principle. Section 5.5 finds camouflage differing by a factor approaching two between relations recorded on the *same* actors, which means our thresholds are per-relation quantities and that a procedure pooling relations is averaging a usable signal with an unusable one.

2.3. Strategic tie formation

The premise that actors form ties instrumentally, and with an eye to how they will be perceived, is established here but thinly populated. Burt et al. [45] show experimentally that brokerage position causes attributions of leadership, and report participants reshaping an assigned network to suit their preferences. Shea et al. [44] come closest to our mechanism: actors anticipating unethical behaviour activate sparser networks that better conceal it, while those who have already acted unethically activate denser ones for support—network structure managed as a function of what the actor wants inferred about it. Barone and Coscia [40] estimate trustworthiness from mutually inconsistent self-reports and find homophily in it, which is both an instance of inferring a latent attribute that actors have an incentive to misreport and a demonstration that the inference can work.

What is absent, so far as we can determine, is the case where the tie portfolio is *engineered* to defeat the inference. We searched the journal's full record and found no treatment of automated or inauthentic accounts; the nearest work concerns misinformation susceptibility in personal networks [51]. We therefore take our terminology from this journal—alter composition, autocorrelation, exposure—and our substantive setting from the detection literature, rather than importing the latter's vocabulary wholesale.

2.4. Degree heterogeneity and the correction it forces

Our edge-endpoint identity requires a degree correction, and the reason is a long-standing concern here. Snijders [47] introduced degree variance as an index of graph heterogeneity; Anderson et al. [46] showed that graph-level indices are systematically confounded by size and density; and Krivitsky et al. [49] exhibit networks with identical, highly skewed degree distributions but entirely different structure, concluding that degree-based accounts cannot by themselves identify structure. Section 5.5 bears this out: the identity requires its degree factor wherever class mean degrees differ, and holds exactly once it is included.

The stochastic blockmodel we build on originates in this journal [48]. Degree correction *within* blockmodels is much less developed here; the one instance we found, Yu et al. [50], monitors node propensity in a dynamic degree-corrected blockmodel in order to detect anomalous nodes, which is close to our setting in both model and purpose.

2.5. Analyses of neighbourhood averaging in the machine-learning literature

The contextual stochastic block model [3] couples a stochastic block model with Gaussian node attributes and has become the standard setting for asking what message passing does to statistical separability. Baranwal et al. [2] established the central positive result: for a two-class CSBM, one graph convolution improves the regime of linear separability by a factor of roughly $1/\sqrt{D}$ in the expected degree D, and the resulting classifier generalises out of distribution to graphs with different edge probabilities.

That analysis assumes the generative process is homophilic—inter-class edges arise from a fixed, non-adversarial probability—and establishes only the positive side. Its follow-ups extend the positive results to multiple layers [17] and characterise the locally Bayes-optimal architecture as one that *interpolates* between an MLP in the low-graph-signal regime and a convolution in the high-signal regime [18]. Our Corollary 8 reduces to the $1/\sqrt{D}$ separability gain of [2] when $\rho = 0$, which calibrates our model against this baseline.

The crossover is known; our contribution is the imbalanced and adversarial case. We state the priority situation plainly, because it determines what this paper does and does not claim. A closed-form threshold at which mean aggregation becomes worse than the graph-free classifier was established by Ma et al. [5, Thm. 2] for the *balanced* symmetric CSBM, and Corollary 8 recovers their condition exactly at $c = 1$ (Remark 9 works the algebra through). Related crossover analyses have since appeared: Luan et al. [15] compute the Bayes error of graph-aware against graph-agnostic classifiers under a homophily parameter with per-class variances and degrees, locating the intersection points directly and criticising the fixed-classifier comparison used in [5]; Mao et al. [16] prove the per-node version, that a GCN beats an MLP on homophilic nodes and loses on heterophilic ones *within the same graph*; and the graph-attention analysis of Fountoulakis et al. [14] both identifies the same signal-deflation factor and proves that attention—the closest analogue to soft edge pruning—is lower-bounded by the better of convolution and the graph-free classifier, while also showing (their Theorems 9 and 13) that no attention mechanism separates intra- from inter-class edges in the hard regime.

Against this background our claim is narrower than "we show when the graph is harmful". It is that the adversarial, class-imbalanced case is not a relabelling of the balanced one: the two classes deflate asymmetrically, which moves the decision boundary and invalidates a fixed-boundary comparison; bot prevalence enters the threshold through $\eta = c\rho$ as a robustness parameter; and the mixture-variance term omitted in [5] shifts the threshold by 11.5%. Read this way, the closest antecedent to Theorem 12 is [14], which gives the qualitative guarantee for attention; we give a closed form for how far a pruner of a given quality moves the threshold.

A parallel literature studies the limits of repeated aggregation. Over-smoothing analyses [7, 8] show node representations converge to an uninformative subspace as depth grows, and spectral accounts [9] characterise graph convolution as a low-pass filter. These results concern degradation with *depth* under a fixed graph. The effect we analyse is orthogonal: a single layer already suffers signal loss when the *topology itself* is adversarial. The two mechanisms compound, but our critical ratio is present at $L = 1$.

2.6. Heterophily in graph learning

The finding that homophily-assuming GNNs degrade on heterophilic graphs motivated a family of architectures that separate ego from neighbour representations, aggregate over higher-order neighbourhoods, or combine low- and high-frequency signals [4, 6]. Ma et al. [5] show in addition that homophily is not strictly necessary: GNNs can succeed on heterophilic graphs when inter-class neighbourhood *distributions* are sufficiently distinguishable.

That observation matters for our setting, and we take a position on it. Under adversarial camouflage the distinguishability Ma et al. rely on is precisely what the adversary is optimising away: a bot that matches the human neighbourhood distribution eliminates the signal their argument requires. The distinction between benign heterophily—structured, and therefore exploitable—and adversarial heterophily, engineered to be unexploitable, is in our view the conceptual point separating bot detection from general heterophilic node classification. It is a distinction about the *provenance* of the inter-class edges, not about the algebra of aggregation, which is why our threshold coincides with theirs in the balanced case while the interpretation of the parameter differs.

2.7. Robust aggregation and adversarial topology

Camouflage is a contamination phenomenon, and two mature literatures have already treated it as one.

The first replaces the mean with a robust estimator. In Byzantine-robust distributed learning a fraction of workers report corrupted updates, and aggregation rules such as Krum [21] and coordinate-wise median or trimmed mean [22] bound the resulting error. The idea has been imported into GNN neighbourhood aggregation directly: Geisler et al. [19] construct an aggregation with the maximal breakdown point of 0.5, so the bias stays bounded while fewer than half of a node's edges are adversarial, and Chen et al. [20] attribute the structural vulnerability of GCNs specifically to the non-robustness of the weighted mean, analysed through the breakdown point.

We therefore avoid the term for our own threshold. A breakdown point asks when an adversary can drive an estimate arbitrarily far under unbounded contamination; $\rho^\star$ is instead a bias–variance crossover under *bounded* contamination—human features are not arbitrary—and marks where the $\sqrt{D}$ variance reduction ceases to pay for the $(1 + c)\rho$ bias. The two quantities answer different questions, and $\rho^\star$ is the weaker of the two.

The second literature attacks the topology. Nettack [26] and Metattack [27] insert or remove edges to flip predictions, and defences such as GNNGuard [28] and Pro-GNN [29] filter them, largely heuristically. Two results in this line bear directly on ours. Zhu et al. [23] prove the bridge our model assumes—that impactful structural attacks on homophilous data necessarily reduce homophily—and derive the ego/neighbour-separation design principle from it. Mujkanovic et al. [24] show empirically that under adaptive attack, defended GNNs can underperform an MLP that ignores the topology outright; our Corollary 8 can be read as a closed-form account of when that should be expected. Certified-robustness work [31] is orthogonal: it certifies that a prediction is unchanged within a perturbation budget, never that the graph has become net-harmful beyond one. Gosch et al. [25] come closest in combining a CSBM, an adversary, and a Bayes-optimal yardstick, but use it to show GNNs are *over*-robust, insensitive past the point of genuine semantic change.

2.8. Camouflage-resistant detectors

CARE-GNN [11] introduced the camouflage framing to GNN-based fraud detection, using a label-aware similarity measure and a reinforcement-learning neighbour selector to filter uninformative edges. Bot-detection work has since adopted the pattern: BotHP [13] separates ego and neighbour encoders to mitigate interaction camouflage, and XHBot [1] combines spectral edge pruning, tri-channel aggregation, and contrastive disentanglement.

These methods are validated by ablation. An ablation establishes that removing a component costs accuracy on a particular benchmark; it does not establish the *range of conditions* under which the component helps, nor whether a multi-channel design is more expressive or merely better parameterised. The present paper supplies exactly those missing statements for the three mechanisms XHBot employs. We analyse XHBot's components because they instantiate the dominant design pattern, not because the analysis is specific to that system: the edge-scoring result applies to any feature-discrepancy pruning rule, and the expressivity result to any aggregator that adds a permutation-invariant non-mean channel.

3. The Camouflage-CSBM

We work in a two-class contextual stochastic block model augmented with an adversarial parameter controlling how aggressively bots attach to humans.

Definition 1 (Camouflage-CSBM)**.** Let $\mathcal{V}$ be a set of N nodes partitioned into bots $\mathcal{B}$ and humans $\mathcal{H}$, with bot prevalence $\pi_B = |\mathcal{B}|/N \in (0, 1)$ and bot-to-human ratio

$c = \pi_B/(1-\pi_B)$. Each node v carries a feature vector

$$\mathbf{x}_v \sim \mathcal{N}(\boldsymbol{\mu}_{y_v}, \sigma^2 I_d), \qquad y_v \in \{\mathrm{B}, \mathrm{H}\}, \tag{1}$$

and we write $\Delta = \boldsymbol{\mu}_{\mathrm{B}} - \boldsymbol{\mu}_{\mathrm{H}}$ for the class-mean gap. Every node has neighbourhood size D. Neighbours are drawn independently: a bot's neighbour is human with probability ρ (the *camouflage ratio*) and a bot otherwise; a human's neighbour is a bot with probability η and a human otherwise.

The two mixing parameters are not free. Counting bot–human edge endpoints from both sides gives $|\mathcal{B}|D\rho = |\mathcal{H}|D\eta$, hence the *edge-balance identity*

$$\eta(\rho) \;=\; \frac{\pi_B}{1-\pi_B}\,\rho \;=\; c\,\rho. \tag{2}$$

Camouflage is therefore doubly harmful: it not only pollutes bot neighbourhoods but necessarily pollutes human neighbourhoods as well, at a rate set by the bot prevalence. Every result below accounts for this coupling; treating η as independent of ρ overstates a detector's robustness.

Assumption 2 (Regularity used by the depth analysis)**.** Definition 1 samples each actor's alters independently, which suffices for the single-layer results. Section 4.5 additionally requires that the tie relation be *symmetric*—so that $v \in \mathcal{N}(u)$ whenever $u \in \mathcal{N}(v)$, which is what makes (19) hold—and that the neighbourhood be locally tree-like with branching D, so that the number of actors at distance j is $N_j = D(D-1)^{j-1}$. Both are standard for sparse networks in which short cycles are rare, and neither is needed before Section 4.5. We also take $\rho \le 1/c$, so that $\eta \le 1$, which holds automatically whenever the strategic class is the minority.

Remark 3 (Modelling choices)**.** Independent neighbour sampling at fixed size D makes the analysis exact and is the discrete analogue of the degree-concentration assumptions used in [2]. Isotropic within-class covariance and a common D are simplifications; Section 6.3 discusses what changes without them. We emphasise that ρ is an *adversarial* parameter chosen by the attacker, not a fixed property of the generative process—this is the substantive difference from the standard CSBM setting.

Aggregation operator. We analyse one round of mean aggregation, the canonical homophilic operator,

$$\mathbf{h}_v \;=\; \frac{1}{D}\sum_{u\in\mathcal{N}(v)} \mathbf{x}_u, \tag{3}$$

which isolates the effect of neighbourhood averaging from the confounds of learned weights and nonlinearity. Because all the quantities we need are governed by the projection onto the discriminant direction, define $\mathbf{w} = \Delta/\|\Delta\|$ and $t_v = \mathbf{w}^\top \mathbf{h}_v$.

Baseline for comparison. The natural point of reference is the graph-free classifier, which uses $\mathbf{x}_v$ directly. For two isotropic Gaussians with equal priors its Bayes error is

$$\varepsilon_{\mathrm{raw}} \;=\; \Phi\!\left(-\frac{\|\Delta\|}{2\sigma}\right), \tag{4}$$

with Φ the standard normal CDF. The question driving Section 4 is when aggregation beats (4) and when it does not.

4. Theory

4.1. Camouflage deflates the signal

We first quantify what aggregation does to the class signal.

Proposition 4 (Signal deflation)**.** Under Definition 1 with mean aggregation (3), the expected post-aggregation separation along the discriminant direction is

$$\delta(\rho) \;=\; \mathbb{E}[t_v \mid y_v = \mathrm{B}] - \mathbb{E}[t_u \mid y_u = \mathrm{H}] \;=\; \big(1-\rho-\eta(\rho)\big)\|\Delta\|, \tag{5}$$

and with the edge-balance identity (2), $\delta(\rho) = \big(1-(1+c)\rho\big)\|\Delta\|$.

Proof. Centre the projection so that $\mathbf{w}^\top\boldsymbol{\mu}_{\mathrm{B}} = +\|\Delta\|/2$ and $\mathbf{w}^\top\boldsymbol{\mu}_{\mathrm{H}} = -\|\Delta\|/2$. For a bot v, each of its D neighbours is human with probability ρ, so by linearity $\mathbb{E}[t_v] = (1-\rho)\frac{\|\Delta\|}{2} + \rho\big(-\frac{\|\Delta\|}{2}\big) = \frac{\|\Delta\|}{2}(1-2\rho)$. Symmetrically $\mathbb{E}[t_u] = -\frac{\|\Delta\|}{2}(1-2\eta)$ for a human u. Subtracting gives $\frac{\|\Delta\|}{2}\big[(1-2\rho)+(1-2\eta)\big] = (1-\rho-\eta)\|\Delta\|$. Substituting (2) yields the second form. □

Two consequences are worth stating. The signal vanishes entirely at $\rho_{\mathrm{null}} = 1/(1+c)$, at which point the two classes have identical post-aggregation means and *no* classifier acting on aggregated features alone can do better than chance. And the deflation is linear in ρ with slope $(1+c)\|\Delta\|$, so higher bot prevalence makes the detector more fragile, not less—a counterintuitive point, since prevalence is usually treated as a class-imbalance nuisance rather than a robustness parameter.

The noise term requires more care than the signal term.

Theorem 5 (Post-aggregation error)**.** Under Definition 1, the projected statistics are asymptotically Gaussian with separation $\delta(\rho)$ from (5) and per-class variances

$$s_{\mathrm{B}}^2 = \frac{\sigma^2 + \rho(1-\rho)\|\Delta\|^2}{D}, \qquad s_{\mathrm{H}}^2 = \frac{\sigma^2 + \eta(1-\eta)\|\Delta\|^2}{D}. \tag{6}$$

Defining the post-aggregation signal-to-noise ratio

$$\mathrm{SNR}(\rho) \;=\; \frac{|\delta(\rho)|}{s_{\mathrm{B}} + s_{\mathrm{H}}}, \tag{7}$$

the balanced error of the variance-equalised linear rule is $\Phi(-\mathrm{SNR}(\rho))$. This is an upper bound on the Bayes error, tight when $s_{\mathrm{B}} = s_{\mathrm{H}}$.

Remark 6 (What this quantity is, and is not)**.** Three clarifications, since the distinction matters for how the comparisons below should be read. (i) When $s_{\mathrm{B}} \neq s_{\mathrm{H}}$ the likelihood ratio of two unequal-variance Gaussians is quadratic, so the Bayes rule is an interval and $\Phi(-\mathrm{SNR})$ is the error of the equal-error-rate *threshold* rule, an upper bound. That bound is loose in one direction, but at finite D it is not the dominant approximation: the central limit theorem is. The aggregate is exactly a mixture of $D+1$ Gaussians, and $\Phi(-\mathrm{SNR})$ *under*-states its error by 41% relative at $\rho = 0.1$, 18% at $\rho = 0.2$ and 6% at $\rho = 0.3$ (Table 1), while the attribute-only baseline is exact because raw attributes are exactly Gaussian. Comparisons therefore *overstate* how much aggregation helps, and are optimistic rather than conservative. The crossover itself is barely affected—exact 0.4642 against 0.4644—but the direction should not be misreported. (ii) The criterion is *balanced* error, weighting the two classes equally, not the prior-weighted risk—a deliberate choice in a paper about class imbalance, where a prior-weighted criterion rewards predicting the majority class and would obscure the effect being measured. At $D = 16$, $\pi_B = 0.3$ the two differ little in practice ($\rho^\star = 0.4644$ balanced against 0.4694 prior-weighted), but the choice should be explicit. (iii) The absolute value in (7) is necessary: without it the expression exceeds 1/2 once ρ passes the point where δ changes sign, which no error probability may do. With it, $\Phi(-\mathrm{SNR})$ correctly records that strongly anti-correlated neighbourhoods are informative again—the effect made explicit in (21).

Proof. Condition on a single neighbour of a bot. Its projected feature equals $\pm\|\Delta\|/2$ according to a Bernoulli(ρ) draw, plus independent $\mathcal{N}(0, \sigma^2)$ noise. By the law of total variance, the mean component contributes $\|\Delta\|^2\rho(1-\rho)$ and the noise contributes σ^2, so each neighbour has variance $\sigma^2 + \rho(1-\rho)\|\Delta\|^2$. The D neighbours are independent, so averaging divides the variance by D, giving s_{B}^2; the human case is identical with η in place of ρ. Asymptotic normality of t_v follows from the central limit theorem as D grows. For two equal-prior Gaussians with common standard deviation s and mean gap δ, the likelihood-ratio test thresholds at the midpoint and attains error $\Phi(-\delta/2s)$, which is (7) when $s_{\mathrm{B}} = s_{\mathrm{H}} = s$. □

Remark 7. The mixture term $\rho(1-\rho)\|\Delta\|^2$ is easy to miss: a calculation that models only the additive noise obtains σ^2/D and therefore *under*-estimates the post-aggregation error, most severely at intermediate ρ where the mixture variance peaks. Section 5 confirms the corrected form matches simulation to four decimal places.

4.2. The critical camouflage ratio

We can now answer the deployment question directly.

Corollary 8 (Critical camouflage ratio)**.** Mean aggregation yields a lower Bayes error than the graph-free classifier (4) if and only if $\mathrm{SNR}(\rho) > \|\Delta\|/(2\sigma)$. Writing $\rho^\star$ for the value of ρ at which equality holds, aggregation helps precisely when $\rho < \rho^\star$. In the noise-dominated regime $\sigma^2 \gg \rho(1-\rho)\|\Delta\|^2$ this takes the closed form

$$\rho^\star = \frac{1 - D^{-1/2}}{1+c}. \tag{8}$$

Proof. The "if and only if" is immediate from monotonicity of Φ. In the noise-dominated regime the mixture terms in (6) are negligible, so $s_{\mathrm{B}} \approx s_{\mathrm{H}} \approx \sigma/\sqrt{D}$ and $\mathrm{SNR}(\rho) \approx (1-\rho-\eta)\|\Delta\|\sqrt{D}/(2\sigma)$. Setting this equal to $\|\Delta\|/(2\sigma)$ gives $(1-\rho-\eta)\sqrt{D} = 1$; substituting $\eta = c\rho$ and solving for ρ yields (8). □

Equation (8) is the paper's most directly actionable result, and it says three things. First, at $\rho = 0$ the condition reduces to $\mathrm{SNR}_{\mathrm{agg}}/\mathrm{SNR}_{\mathrm{raw}} = \sqrt{D}$: aggregation improves separability by $\sqrt{D}$, recovering the $1/\sqrt{D}$ gain established by Baranwal et al. [2] and confirming our model is calibrated against the known homophilic baseline. Second, $\rho^\star$ increases with D, so detectors operating on densely connected accounts tolerate more camouflage—the graph is a genuine asset for hub accounts long after it has become a liability for sparse ones. Third, $\rho^\star$ decreases in c: as bots become more prevalent, the camouflage budget they need in order to break the detector shrinks.

Remark 9 (Relation to Ma et al.)**.** Equation (8) is not new in the balanced case, and we want to be explicit about what is and is not ours. For a symmetric CSBM with intra- and inter-class edge probabilities p and q, Ma et al. [5, Thm. 2] prove that a node is better classified after aggregation than before whenever $\deg(i) > (p+q)^2/(p-q)^2$. Their inter-class neighbour fraction is $\rho = q/(p+q)$, so $(p-q)/(p+q) = 1-2\rho$ and their condition rearranges to $\sqrt{D}\,(1-2\rho) > 1$, i.e. $\rho < (1-D^{-1/2})/2$. This is exactly (8) at $c = 1$. A closed-form threshold for when mean aggregation is worse than discarding the graph was therefore established at ICLR 2022, and our corollary is its extension to $c \neq 1$.

What the extension adds is not merely a symbol. Three things change when the classes are imbalanced. (i) The two classes deflate by *different* amounts, ρ for bots and $\eta = c\rho$ for humans, so the post-aggregation class midpoint *moves*; a comparison against a decision boundary held fixed at the raw-feature midpoint—as in [5]—is then no longer the right one, which is precisely the criticism Luan et al. [15] raise against that analysis. Our Theorem 5 compares Bayes errors instead, and is insensitive to this. (ii) The edge-balance identity $\eta = c\rho$ in (2) is what couples the two, and it is what makes bot prevalence a robustness parameter rather than a training-time nuisance. (iii) The mixture-variance term $\rho(1-\rho)\|\Delta\|^2$ is absent from the covariance $I/\sqrt{\deg(i)}$ reported in [5]; including it moves $\rho^\star$ from 0.53 to 0.46 in the configuration of Remark 10, an 11.5% correction to the threshold.

Remark 10 (Practical reading)**.** For a detector with typical $D = 16$ and bot prevalence $\pi_B = 0.3$ ($c \approx 0.43$), (8) gives $\rho^\star \approx 0.53$ in the noise-dominated regime and 0.46 once the mixture variance is included. A bot that directs roughly half its edges at genuine users has therefore neutralised the graph entirely. This is not an extreme adversary; it is a modest one, which is why architectural mitigation is necessary rather than optional.

4.3. Spectral edge pruning provably extends the operating regime

SGTR [1] scores each edge by a learned function of the feature discrepancy $(\mathbf{x}_u - \mathbf{x}_v)^2$, motivated by the per-edge decomposition of the Dirichlet energy $\mathbf{f}^\top L\mathbf{f} = \sum_{(u,v)\in\mathcal{E}}(f_u - f_v)^2$. The following lemma shows this quantity is the right statistic: it separates camouflage from genuine edges by exactly the squared class gap, independently of the feature dimension.

Lemma 11 (Energy separability)**.** Under Definition 1, for a within-class edge (u, v) and a cross-class (camouflage) edge (u', v'),

$$\mathbb{E}\left\|\mathbf{x}_u - \mathbf{x}_v\right\|^2 = 2d\sigma^2, \qquad \mathbb{E}\left\|\mathbf{x}_{u'} - \mathbf{x}_{v'}\right\|^2 = \|\Delta\|^2 + 2d\sigma^2, \tag{9}$$

so the expected energy gap is exactly $\|\Delta\|^2$.

Proof. For a within-class edge both endpoints share a mean, so $\mathbf{x}_u - \mathbf{x}_v \sim \mathcal{N}(0, 2\sigma^2 I_d)$ and the expected squared norm is $2d\sigma^2$. For a cross-class edge $\mathbf{x}_{u'} - \mathbf{x}_{v'} \sim \mathcal{N}(\Delta, 2\sigma^2 I_d)$, and $\mathbb{E}\|Z\|^2 = \|\mathbb{E}Z\|^2 + \operatorname{tr}\operatorname{Cov}(Z)$ gives $\|\Delta\|^2 + 2d\sigma^2$. □

The gap is a *location* shift that does not grow with d, whereas the common term $2d\sigma^2$ does; the score is therefore informative but increasingly noisy in high dimension, which is precisely why XHBot learns a projection rather than using the raw quadratic form. We now quantify the benefit of acting on this score.

Theorem 12 (SGTR raises the tolerated camouflage)**.** Suppose soft pruning multiplies camouflage edge weights by $(1-\bar{e}_c)$ and within-class edge weights by $(1-\bar{e}_w)$ in expectation, with $\bar{e}_c > \bar{e}_w$. Then weighted aggregation behaves as unweighted aggregation at the *effective* camouflage ratio

$$\rho_{\text{eff}}(\rho) = \frac{\rho(1-\bar{e}_c)}{\rho(1-\bar{e}_c) + (1-\rho)(1-\bar{e}_w)} < \rho. \tag{10}$$

Consequently, with $\bar{e}_w = 0$, the detector tolerates true camouflage up to

$$\rho^\star_{\text{SGTR}} = \frac{\rho^\star}{\rho^\star + (1-\bar{e}_c)(1-\rho^\star)} \geq \rho^\star, \tag{11}$$

with equality if and only if $\bar{e}_c = 0$.

Equation (11) accounts only for the change in neighbourhood *composition*. It therefore appears to imply $\rho^\star_{\text{SGTR}} \to 1$ as $\bar{e}_c \to 1$: that a sufficiently good pruner tolerates any camouflage whatsoever. That conclusion is false, and Section 4.4 shows why. Down-weighting edges non-uniformly also reduces the *effective degree*, and hence forfeits part of the $\sqrt{D}$ variance reduction that made aggregation worthwhile in the first place. Once that cost is included the naive reading fails, though not in the direction it first appears: Proposition 15 shows there is no hard ceiling at all, because a perfect pruner leaves the aggregate unbiased. What changes is the *rate* at which the advantage thins out, and the fact that the operating point must be chosen on a detection trade-off rather than by pushing $\bar{e}_c \to 1$.

Proof. Weighted mean aggregation computes $\sum_e w_e \mathbf{x}_{u_e} / \sum_e w_e$. In expectation a fraction ρ of a bot's edges are camouflage edges carrying weight $(1-\bar{e}_c)$ and $(1-\rho)$ are within-class edges carrying $(1-\bar{e}_w)$; normalising gives the weight share (10) on human neighbours, which is exactly the role ρ plays in Proposition 4. The inequality $\rho_{\text{eff}} < \rho$ follows since $\bar{e}_c > \bar{e}_w$ makes the numerator shrink faster than the denominator. For (11), set $\rho_{\text{eff}}(\rho) = \rho^\star$ with $\bar{e}_w = 0$ and solve: writing $\beta = 1 - \bar{e}_c$, the equation $\rho\beta = \rho^\star(\rho\beta + 1 - \rho)$ rearranges to $\rho[\beta(1-\rho^\star) + \rho^\star] = \rho^\star$, giving the stated form. Monotonicity in $\bar{e}_c$ and the limits are immediate. □

Theorem 12 converts an ablation delta into a statement about the operating regime: a pruning module that attenuates camouflage edges by 75% on average lifts the tolerated camouflage from 0.53 to 0.82 at $D = 16$, $\pi_B = 0.3$. It also clarifies what pruning cannot do. Because $\rho_{\text{eff}} \to 1$ as $\rho \to 1$ for any fixed $\bar{e}_c < 1$, no pruning rule with imperfect discrimination defeats a maximally camouflaged adversary; SGTR buys operating range, not immunity.

Remark 13 (What is new here, and what is not)**.** The *mechanism* in (10)—that edge reweighting acts by rescaling the effective inter-class edge share, so that weights enter only through the products of weight and edge rate—is not new. Weiss [30, Thm. 2] proves the corresponding statement for cost-sensitive aggregation with within- and cross-class weights $w_+ > w_-$, and shows the class-mean direction is preserved iff $w_+/w_- > q/p$; our (10) is that identity with $w_+ = 1 - \bar{e}_w$ and $w_- = 1 - \bar{e}_c$. The observation that only a perfect discriminator tolerates arbitrary heterophily is the contrapositive of the same condition. The related guarantee for graph attention in [14] is qualitative but similar in spirit.

What (11) adds is the *inversion*: composing that rescaling with the finite-D threshold of Corollary 8 to obtain a bound on the *adversary's* budget as an explicit function of pruner quality $\bar{e}_c$. This is a short derivation, and we present it as a corollary of two known ingredients rather than as an independent discovery. Its value is operational: $\bar{e}_c$ is measurable for a deployed pruner, so (11) turns a component ablation into a quantitative operating-range claim. We note also that [30] works in the dense regime where degree effects are absorbed into $O(1/n)$, which removes the $D^{-1/2}$ term that our threshold depends on; the finite-D setting is therefore necessary for a statement of this form.

4.4. What pruning quality is achievable, and what it costs

Theorem 12 takes the pruner's quality $\bar{e}_c$ as given. Two things are then missing: whether a given $\bar{e}_c$ is attainable at all by a feature-based score, and what is given up in exchange. We supply both, which removes the free parameter and corrects the limit noted above.

Only the retained-weight ratio matters. Write $a = 1 - \bar{e}_c$ and $b = 1 - \bar{e}_w$ for the surviving weight on camouflage and within-class edges. Weighted mean aggregation is invariant to a common rescaling of all weights, so a and b enter only through the ratio

$$\theta \;=\; a/b \;\in\; (0, 1], \tag{12}$$

which we call the *retained-weight ratio*; $\theta = 1$ is no pruning and $\theta = 0$ is perfect pruning. This is the same quantity that governs the cost-sensitive condition $w_+/w_- > q/p$ of [30]. In terms of θ, the effective camouflage ratio of Theorem 12 is $\rho_{\text{eff}} = \rho\theta/(\rho\theta + 1 - \rho)$.

Proposition 14 (Pruning forfeits effective degree)**.** A weighted mean of D unit-variance terms has the variance of an unweighted mean of

$$D_{\text{eff}} \;=\; D\,\frac{\big[(1-\rho)+\rho\theta\big]^2}{(1-\rho)+\rho\theta^2} \;\le\; D \tag{13}$$

terms, with equality if and only if $\theta = 1$.

Pruning acts on *both* ends of a camouflage edge, so the human side is rescaled too and its cross-class share becomes $\eta_{\text{eff}} = \eta\theta/(\eta\theta + 1 - \eta)$ with $\eta = c\rho$. Note that $\eta_{\text{eff}} \neq c\,\rho_{\text{eff}}$ unless $c = 1$ or $\theta = 1$: the edge-balance identity holds for the *raw* ratios and is not preserved by pruning. Writing $\lambda_{\text{eff}} = 1 - \rho_{\text{eff}} - \eta_{\text{eff}}$ and $D^B_{\text{eff}}, D^H_{\text{eff}}$ for (13) evaluated at ρ and at η respectively, pruned aggregation beats the graph-free classifier if and only if

$$\frac{|\lambda_{\text{eff}}|}{\sqrt{\tfrac{1}{2}\big(1/D^B_{\text{eff}} + 1/D^H_{\text{eff}}\big)}} \;>\; 1. \tag{14}$$

Proof. The variance of $\sum_e w_e z_e / \sum_e w_e$ for independent unit-variance z_e is $\sum_e w_e^2/(\sum_e w_e)^2$, so the variance-equivalent count is $(\sum_e w_e)^2/\sum_e w_e^2$. Substituting ρD edges of weight a and $(1-\rho)D$ of weight b, and cancelling b^2, gives (13); the inequality is Cauchy–Schwarz, with equality iff the weights are constant. Combining with Proposition 4 applied at ρ_{eff} yields (14). □

This resolves a degeneracy that (11) alone does not. Uniform attenuation ($\theta = 1$, any common scale) leaves both ρ_{eff} and D_{eff} unchanged and so achieves exactly nothing, as it must; and aggressive pruning is now penalised rather than free.

Proposition 15 (A perfect pruner never loses, but its benefit thins out)**.** Under perfect pruning every retained neighbour shares the node's label, so the retained count is $K \sim \mathrm{Bin}(D, 1-\rho)$ and the aggregate is *unbiased* with variance σ^2/K. Consequently, for every $\rho < 1$, pruned aggregation is never worse than the graph-free classifier: it is strictly better on the event $\{K \ge 2\}$, and ties when $K \in \{0, 1\}$. The fraction of *strategic actors* that strictly benefit is

$$\Pr[K \ge 2] \;=\; 1 - \rho^D - D(1-\rho)\rho^{D-1}, \tag{15}$$

and the variance-equivalent degree is $1/\mathbb{E}[1/K \mid K \ge 1]$, not $\mathbb{E}[K]$.

Proof. With all camouflage edges removed the surviving neighbours are same-class, so their mean has the node's own class mean as its expectation: the bias is zero regardless of ρ. Conditional on $K = k \ge 1$ the variance is $\sigma^2/k \le \sigma^2$, with strict inequality iff $k \ge 2$; nodes with $K = 0$ have no neighbourhood and fall back on their own features, tying exactly. Since K is Binomial, $\Pr[K \le 1]$ is the sum of its first two terms, giving (15). That the variance-equivalent degree is the harmonic rather than arithmetic mean follows because the variance of a mean of K terms is $\sigma^2\mathbb{E}[1/K]$, and $\mathbb{E}[1/K] \ge 1/\mathbb{E}[K]$ by Jensen. □

Remark 16 (A tempting but incorrect ceiling)**.** Substituting the *mean* retained count $D(1-\rho)$ into (13) yields $D_{\text{eff}} = D(1-\rho)$ and hence the clean-looking conclusion that even perfect pruning requires $\rho < 1 - 1/D$. That is wrong, and instructively so. The substitution replaces a random K by its mean in a quantity that depends on

$1/K$, and it is applied in precisely the regime where the replacement is least defensible: at $\rho = 1 - 1/D$ and $D = 16$ we have $\mathbb{E}[K] = 1.00$ exactly, yet $1/\mathbb{E}[1/K \mid K \geq 1] = 1.29$, and 36% of nodes have $K = 0$. Proposition 15 shows the qualitative conclusion inverts: there is no hard ceiling. What does happen as $\rho \to 1$ is that the *proportion* of nodes for which the graph carries any information at all decays; a useful summary is the point at which half of the strategic actors still benefit, which is $\rho = 0.897$ at $D = 16$ and 0.974 at $D = 64$.

Achievable θ. It remains to bound θ from below. SGTR scores an edge by the feature discrepancy $s_{uv} = \|\mathbf{x}_u - \mathbf{x}_v\|^2$, whose law under the CSBM is exactly

$$s \sim 2\sigma^2\chi^2_d \quad \text{(within class)}, \qquad s \sim 2\sigma^2\chi^2_d\left(\tfrac{r^2}{2}\right) \quad \text{(camouflage)}, \tag{16}$$

with $r = \|\Delta\|/\sigma$ and $\chi^2_d(\lambda)$ the noncentral chi-square. Any threshold rule therefore has an operating point $(\bar{e}_w, \bar{e}_c)$ on the ROC of (16), and θ is the ratio of its complements. In the large-d Gaussian approximation the discriminability index is

$$\kappa = \frac{\|\Delta\|^2}{2\sigma^2\sqrt{2d}} = \frac{r^2}{2\sqrt{2d}}, \tag{17}$$

so $\bar{e}_c \approx \Phi(\Phi^{-1}(\bar{e}_w) + \kappa)$.

Two consequences follow. First, $\bar{e}_w \to 0$ forces $\bar{e}_c \to 0$: there is no threshold that removes camouflage edges without also removing genuine ones, so the idealisation $\bar{e}_w = 0$ used in (11) is not attainable and the joint form is required. Second, because κ scales as $d^{-1/2}$, the raw discrepancy score *degrades* as the feature dimension grows—the two score distributions concentrate and overlap. This is the formal content of the remark that motivates learning a projection: mapping to $k \ll d$ dimensions while preserving $\|\Delta\|$ multiplies κ by $\sqrt{d/k}$.

The soft-weight and hard-threshold models must not be mixed. Equation (13) assigns every camouflage edge the same weight a and every genuine edge the same b, so it is invariant to a common rescaling and depends only on θ. A *threshold* rule is different: it deletes edges outright, so the neighbourhood shrinks in absolute terms and the effective degree is

$$D^{\text{hard}}_{\text{eff}} = D\Big[(1-\rho)(1-\bar{e}_w) + \rho\,(1-\bar{e}_c)\Big], \tag{18}$$

which collapses as $\bar{e}_w \to 1$. Optimising the operating point under the scale-invariant model (13) while reading $(\bar{e}_w, \bar{e}_c)$ off a threshold ROC is therefore inconsistent, and the inconsistency is not benign: it rewards discarding almost every genuine edge at no modelled cost, so the apparent optimum runs to $\bar{e}_w \to 1$ and no interior solution exists. We therefore optimise (18) together with the ROC of (16), which has a genuine interior optimum and is stable under enlargement of the search domain. The resulting tolerance depends only on (D, c, r, d), all measurable. Section 5 reports the values, which are considerably less flattering to raw-discrepancy pruning than the inconsistent pairing suggests.

4.5. Depth

Everything so far concerns a single layer. Since deployed detectors stack two or more, we record how the picture changes. The analysis requires care, because the obvious recursion—each layer multiplying the separation by λ—is *not* what mean aggregation does.

Mean aggregation backtracks. On a D-regular tree, two rounds of neighbourhood averaging give

$$\mathbf{h}^{(2)}_v = \frac{1}{D^2}\sum_{u\in\mathcal{N}(v)}\sum_{w\in\mathcal{N}(u)}\mathbf{x}_w = \frac{1}{D}\mathbf{x}_v + \Big(1 - \frac{1}{D}\Big)\bar{\mathbf{x}}_{\text{2-hop}}, \tag{19}$$

because $v \in \mathcal{N}(u)$ for every $u \in \mathcal{N}(v)$: the walk can step back. The ego node therefore receives weight exactly $1/D$, and it carries *undeflated* signal. A recursion that propagates only outwards describes the *non-backtracking* operator, not L rounds of mean aggregation, and the tree-like assumption does not license the substitution—the backtracking term is present on an infinite tree. We therefore work with the true operator $P = D^{-1}A$.

Theorem 17 (Depth). Let $q^{(L)}_j$ be the probability that an L-step simple random walk from v on the D-regular tree ends at distance j, and $N_j = D(D-1)^{j-1}$ ($N_0 = 1$) the number of nodes at that distance. Then, in the noise-dominated regime,

$$\frac{\mathrm{SNR}_L}{\mathrm{SNR}_{\text{raw}}} = \frac{\left|\sum_j q^{(L)}_j \lambda^j\right|}{\sqrt{\sum_j \left(q^{(L)}_j\right)^2 / N_j}}, \qquad \lambda = 1 - (1+c)\rho. \tag{20}$$

The label sequence along the walk is a two-state Markov chain with second eigenvalue λ, which is why the depth-j contribution is deflated by λ^j rather than by λ^L.

Proof. Deferred to Appendix A.2. □

At $L = 1$, $q_1 = 1$ and $N_1 = D$, so (20) reduces to $|\lambda|\sqrt{D}$ and recovers Corollary 8. For $L \geq 2$ it does not factorise, and two consequences follow.

First, **the crossover is not depth-independent**. Solving (20) $= 1$ at $D = 16$, $\pi_B = 0.3$ gives $\rho^\star_1 = 0.525$, $\rho^\star_2 = 0.587$, $\rho^\star_3 = 0.553$, $\rho^\star_4 = 0.576$, $\rho^\star_5 = 0.557$. Every value for $L \geq 2$ exceeds the single-layer threshold, so **depth buys tolerance to camouflage**.

The mechanism is that backtracking retains weight at distances $j < L$, which are deflated by λ^j rather than by λ^L; the ego term $j = 0$ of (19) is the extreme case, being undeflated entirely. It is worth being precise about this, because the ego term is *not* present at every layer: the tree is bipartite, so $q_0^{(L)} = 0$ for all odd L. At $L = 3$ the ego weight is zero, yet $\rho_3^\star = 0.553$ still exceeds $\rho_1^\star$—there the gain comes from mass returned to distance 1 ($q_1^{(3)} = 0.121$). The oscillation in L is this parity: even L admits an ego term, odd L does not.

Two qualifications matter. Deleting the ego term at $L = 2$ and renormalising drops the crossover from 0.587 to 0.522, so at even depths it does carry most of the effect. And the benefit is specific to the camouflaged regime: backtracking *costs* separability when camouflage is absent, since the retained near-origin mass is over-weighted noise. At $\rho = 0$ the SNR ratios are $4.00, 11.49, 29.73, 72.91$ for $L = 1, \dots, 4$ against $4.00, 15.49, 60.0, 232.4$ for a non-backtracking walk. Depth under backtracking therefore trades peak separability at low ρ for a higher crossover—a bias–variance trade rather than a free gain, and one that favours architectures able to tune the ego weight rather than inherit it at $1/D$ [4, 23].

Second, the harmful region is *bounded above*. Only $|\lambda|$ enters, so at $L = 1$ aggregation is harmful precisely on the window

$$\rho \in \left(\frac{1 - D^{-1/2}}{1 + c}, \ \frac{1 + D^{-1/2}}{1 + c} \right) \cap (0, 1], \qquad (21)$$

and helps again above it: a bot with $\rho \to 1$ attaches almost exclusively to humans, so its neighbourhood is strongly *anti*-correlated with its label and informative once more, with the orientation reversed. We claim no novelty for this. It is the second root of the same inequality $\sqrt{D}\,|1 - 2\rho| > 1$ that Remark 9 attributes to Ma et al. [5], whose symmetry in $|p - q|$ those authors discuss directly, and the resulting "intermediate heterophily is worst" phenomenon is the mid-homophily pitfall named by Luan et al. [15]. We record it because it identifies the dangerous adversarial strategy as *intermediate* rather than maximal camouflage, and because the upper edge exceeds 1—so that the window degenerates to a one-sided threshold and no "helps again" regime exists—whenever $c < D^{-1/2}$, which is the case for all of the real graphs measured in Section 5.5.

4.6. Implications for automated detection

The three results above are statements about network structure, but they bear directly on how automated accounts are detected in practice, and it is worth making the connection explicit because the current literature addresses these mechanisms architecturally rather than by measurement.

Detectors built on neighbourhood aggregation now routinely include a module that down-weights or removes suspicious ties, a second aggregation channel for dissimilar neighbours, or an explicit separation of an actor's own attributes from its neighbours' [1, 11, 13]. Each of these is justified in the literature by an ablation: removing the module costs accuracy on a benchmark. Our analysis supplies the missing account of *why* they help and, more usefully, *when*.

Tie-discarding modules are the case analysed in Section 4.4, and the analysis is less encouraging than the ablations suggest: discarding ties trades bias against the number of ties left to average over, so its benefit is bounded by the quality of the underlying suspicion score, and on the networks we measure a score based on attribute discrepancy alone is far too weak to restore inference (Section 5.5). Separating an actor's own attributes from its neighbours' is, by Theorem 17, a way of making explicit and tunable a term that iterated averaging already supplies at the fixed weight $1/D$—which explains why the device helps most in exactly the regime, low degree and high camouflage, where that fixed weight is least appropriate. And the finding that camouflage differs by a factor approaching two across relation types within one population implies that a detector pooling relations into a single adjacency is averaging a usable signal with an unusable one; our thresholds apply per relation, and suggest treating relation selection as a measurement question prior to model choice.

We draw one broader methodological conclusion. Because ρ, c, D and the attribute separation are all estimable from data a practitioner already has, the question "is relational information worth using here?" can be answered before fitting any model, and per relation rather than for a network as a whole. That is a more informative starting point than an ablation, which reports only that a component helped on one benchmark without indicating the conditions under which it would stop helping.

5. Numerical Validation

Every closed form in Section 4 is checked against Monte-Carlo simulation of Definition 1. Our purpose here is verification of the analysis, not a performance claim: the results below establish that the derivations are correct, and say nothing on their own about accuracy on real bot graphs. Section 6.3 addresses that gap directly.

Protocol. Unless stated otherwise we use $\pi_B = 0.3$ ($c \approx 0.4286$), $\|\Delta\| = 2$, $\sigma = 1$, and $D = 16$, drawing 4×10^5 nodes per class per configuration (1.5×10^6 for the crossover sweep). Errors are computed by exhaustive threshold search over the empirical distributions, so the

Table 1. Theory versus simulation for signal deflation and balanced error ($\pi_B = 0.3$, $\|\Delta\| = 2$, $\sigma = 1$, $D = 16$).

		separation δ		s_B		error
ρ	η	theory	sim.	theory	sim.	thy / sim.
0.00	0.000	2.0000	2.0000	0.2500	0.2499	0.00003 / 0.00003
0.10	0.043	1.7143	1.7142	0.2915	0.2918	0.00113 / 0.00160
0.20	0.086	1.4286	1.4283	0.3202	0.3204	0.00925 / 0.01095
0.30	0.129	1.1429	1.1441	0.3391	0.3391	0.03698 / 0.03895
0.40	0.171	0.8571	0.8586	0.3500	0.3499	0.09785 / 0.09898
0.50	0.214	0.5714	0.5696	0.3536	0.3528	0.19898 / 0.19974
0.60	0.257	0.2857	0.2864	0.3500	0.3500	0.33734 / 0.33680

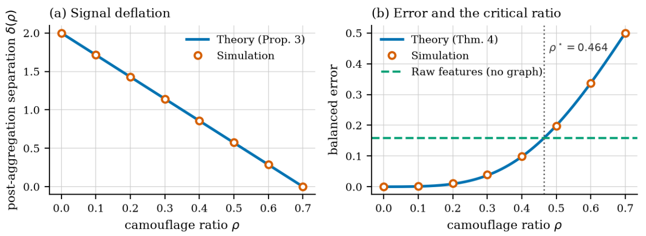

Figure 1. Theory versus simulation. (a) Post-aggregation separation decays linearly in ρ exactly as Proposition 4 predicts. (b) Balanced error against the attribute-only baseline; the predicted critical ratio $\rho^\star = 0.464$ coincides with the crossing point.

reported figures are attained minima rather than plug-in estimates. All code is released with the paper and runs in a few minutes on CPU.

5.1. Deflation and error

Table 1 compares Proposition 4 and Theorem 5 against simulation. Separation matches to within 1.2×10^{-3} and the standard deviation—including the mixture term that a naive calculation omits—to within 4×10^{-4} across the whole range. Figure 1 shows the same agreement graphically.

5.2. The critical ratio

The central prediction is Corollary 8. Solving $\mathrm{SNR}(\rho) = \|\Delta\|/(2\sigma)$ numerically at $D = 16$ gives $\rho^\star = 0.4644$. Sweeping ρ in steps of 0.01 with 1.5×10^6 samples per point, aggregation beats raw features at $\rho = 0.46$ ($\varepsilon = 0.1545$ against $\varepsilon_{\mathrm{raw}} = 0.1587$) and loses at $\rho = 0.47$ ($\varepsilon = 0.1648$). The measured crossover therefore lies in $[0.46, 0.47]$, containing the predicted value.

Table 2 separates the two forms of the corollary. The exact crossover differs from the closed form (8) at moderate SNR, as expected, since (8) drops the mixture variance. Re-running the exact solver in the noise-dominated regime it was derived for ($\|\Delta\| = 0.2$, $\sigma = 1$) recovers the closed form to three decimal places at every degree, confirming that the discrepancy is the stated approximation and not an error.

Table 2. Critical camouflage ratio: closed form (8) against the exact SNR crossover, at moderate and low SNR.

D	closed form (8)	exact ($\|\Delta\|/\sigma = 2$)	exact (low SNR)
4	0.3500	0.2604	0.3488
9	0.4667	0.3913	0.4657
16	0.5250	0.4644	0.5243
25	0.5600	0.5102	0.5594
64	0.6125	0.5807	0.6121

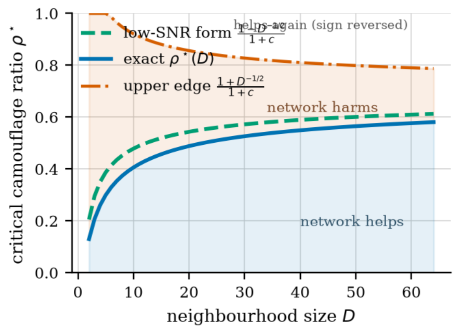

Figure 2. Critical camouflage ratio against neighbourhood size. Below the lower curve, averaging over alters improves on attributes alone; between the curves it is actively harmful; above the upper edge $(1 + D^{-1/2})/(1 + c)$ the neighbourhood is strongly anti-correlated with the label and informative again with the orientation reversed. The closed form is the noise-dominated limit of the exact crossover, and the upper edge exceeds 1—so that no reversal regime exists—whenever $c < D^{-1/2}$.

Figure 2 plots both curves against degree. The tolerated camouflage rises steeply for sparsely connected accounts and flattens as D grows, which has a concrete operational reading: the accounts most vulnerable to camouflage are those with few connections, and a deployment that aggregates over small neighbourhoods is fragile in exactly the regime where bots are cheapest to create.

5.3. SGTR

Lemma 11 predicts expected squared discrepancies of $2d\sigma^2$ and $\|\Delta\|^2 + 2d\sigma^2$. At $d = 8$, $\sigma = 1$, $\|\Delta\| = 2$ these are 16 and 20; simulation over 2×10^5 edge samples gives 16.005 and 20.005, an error below 0.03%.

For Theorem 12 we verify the closed form (11) by round-trip: substituting $\rho^\star_{\mathrm{SGTR}}$ back into ρ_{eff} must return $\rho^\star$. The maximum absolute deviation over $\bar{e}_c \in \{0.25, 0.5, 0.75, 0.9\}$ is 1.1×10^{-16}, i.e. machine precision. Table 3 reports the resulting tolerance gains, and Figure 3 plots the full curve.

Table 3. Effect of SGTR attenuation on tolerated camouflage under the composition-only form (11) ($D = 16$, $\pi_B = 0.3$, baseline $\rho^\star = 0.525$). These values ignore both the effective-degree cost and the correct η_{eff}; Section 5.4 gives the figures we stand behind. The table is retained only to show the size of the two corrections.

$\bar{e}_c$	ρ_{eff} at $\rho = 0.4$	$\rho^\star_{\text{SGTR}}$
0.00	0.4000	0.5250
0.25	0.3333	0.5957
0.50	0.2500	0.6885
0.75	0.1429	0.8155
0.90	0.0625	0.9170

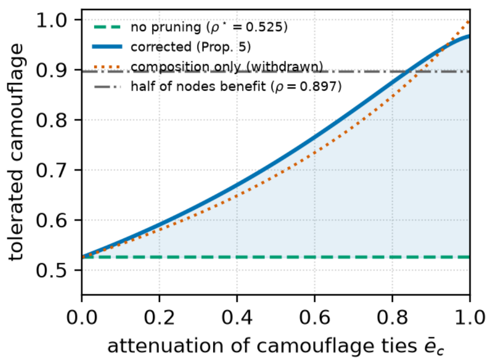

Figure 3. Tolerated camouflage against the attenuation applied to camouflage ties. The solid curve is the corrected analysis (Proposition 14, with η_{eff} as in (14)); the dotted curve is the composition-only form (11). The correction is *upward* over almost the whole range—treating the human side correctly matters more than the effective-degree penalty—but it saturates at 0.967 rather than reaching 1, because pruning cannot manufacture ties that camouflage has removed. The dash-dotted line marks where half of the *strategic actors* still retain enough ties to benefit (Proposition 15); genuine actors are barely affected.

5.4. The extensions: achievable pruning and depth

The effective-degree formula (13) matches Monte-Carlo variance to within 0.5% for $\theta \geq 0.5$. At $\theta = 0$ it is off by 7.3% (8.00 predicted against 7.41 measured at $\rho = 0.5$, $D = 16$), and the cause is not sampling: it is Jensen's inequality on $\mathbb{E}[1/K]$ for the random retained count K, exactly the effect quantified in Remark 16. Under perfect pruning the variance-equivalent degree is 1.29 rather than the 1.00 that $\mathbb{E}[K]$ predicts at $\rho = 1 - 1/D$, $D = 16$, and 35.6% of nodes retain no neighbour at all; the proportion still benefiting falls to one half at $\rho = 0.897$ ($D = 16$) and 0.974 ($D = 64$).

Correcting η_{eff} as in (14) raises the tolerances relative to the $c\,\rho_{\text{eff}}$ substitution: at $D = 16$, $\pi_B = 0.3$ we obtain 0.609, 0.715, 0.847 and 0.928 for $\bar{e}_c = 0.25, 0.5, 0.75, 0.9$, against 0.594, 0.679, 0.788 and 0.869 under the incorrect form. The error was therefore pessimistic, and by up to six percentage points.

For depth, `theory/corrections.py` evaluates (20) directly. The weight profile $\sum_j (q_j^{(L)})^2 / N_j$ exceeds the non-backtracking value D^{-L} by factors of 1.94, 4.63, 12.3 and 35.0 at $L = 2, \ldots, 5$, which is the quantitative size of the term that (19) restores. The resulting crossovers are $\rho_1^\star = 0.5250$, $\rho_2^\star = 0.5868$, $\rho_3^\star = 0.5528$, $\rho_4^\star = 0.5756$ and $\rho_5^\star = 0.5567$: every multi-layer value lies above the single-layer one, so depth increases the tolerated camouflage, and the crossover is *not* depth-independent.

5.5. Measuring the camouflage ratio on real graphs

The theory is parameterised by ρ, which has so far been a modelling device. We now measure it. TwiBot-20 and TwiBot-22 require author approval for access, which we did not have. We therefore use **MGTAB** [10], an openly distributed Twitter dataset of 10,199 expert-annotated accounts (2,748 automated, 7,451 genuine; $c = 0.369$) in which *every* account carries a label, so that all ties are between labelled actors and the measurement is not diluted by unlabelled alters. MGTAB records seven relations on the same population, which lets us separate the social layer (following, friendship, mention, reply, quote) from a content-coordination layer (shared URLs, shared hashtags). For comparison we also report the two review networks of Dou et al. [11]; these are not social ties, and we treat them only as a contrast case.

A negative result on the canonical benchmark. It is worth recording why we do not use cresci-2015 or cresci-2017, the longest-standing bot benchmarks. In the openly distributed preprocessed release, cresci-2017 contains *no* interpersonal ties at all: all 6,637,615 of its edges are account-to-tweet postings. cresci-2015 does contain following and friendship ties, but only 0.41% of the 1.29 million accounts they span are labelled, and the labelled-to-labelled subgraph reduces to 4,275 ties with the mixing pattern 3,823 genuine–genuine, 373 automated–automated and just **79** cross-class. Only 698 of 3,351 automated accounts (20.8%) have any labelled tie, the median such account has degree 0, and the median camouflage ratio is therefore 0. A quantity resting on 79 ties cannot support a distributional claim, and we obtained the same counts from a second, independently preprocessed release of the same data, so this is a property of the dataset rather than of one pipeline. We report it because it explains a gap in the literature: the benchmark most used for bot detection cannot be used to measure the structural mechanism the field attributes to bots.

For each relation we compute ρ as the mean over automated accounts of the fraction of their alters that

Table 4. Camouflage measured on MGTAB (10,199 Twitter accounts, all labelled; $c = 0.369$). $\hat{\rho}$ is the mean genuine share of an automated account's alters, $\tilde{\rho}$ its median, $\tilde{D}$ the median degree over automated accounts having at least one tie of that relation. The last two columns evaluate the *exact* criterion of Theorem 5 at $r = 2.32$ (Section 5.6), not the noise-dominated window (21): ρ_{cross} is the crossover and the final column is $\text{SNR}_{\text{agg}}/\text{SNR}_{\text{raw}}$ at the measured $\hat{\rho}$, so values below 1 mean the relation is worse than using attributes alone.

Relation	ties	$\hat{\rho}$	$\tilde{\rho}$	$\hat{\eta}$	$\tilde{D}$	ρ_{cross}	$\frac{\text{SNR}_{\text{agg}}}{\text{SNR}_{\text{raw}}}$
social layer							
following	231,829	0.969	1.00	0.041	4	0.252	0.51
friendship	327,035	0.989	1.00	0.033	8	0.372	0.79
mention	107,794	0.994	1.00	0.022	4	0.252	0.57
quote	20,536	0.999	1.00	0.014	2	0.121	0.42
reply	20,700	0.991	1.00	0.009	1	—	0.28
union	331,251	0.989	1.00	0.032	8	0.372	0.79
content-coordination layer							
shared URL	263,800	0.595	0.62	0.195	12	0.432	0.44
shared hashtag	300,000	0.627	0.65	0.279	53	0.585	0.71

are genuine, and η symmetrically. Table 4 reports the result.

5.6. Estimating the attribute separation

The criterion depends on $r = \|\Delta\|/\sigma$, and this quantity is easy to overstate. The obvious plug-in—$\|\Delta\|$ from the class-mean difference and σ from the mean per-dimension variance—gives $r = 13.0$ on MGTAB's 788 attributes. That figure is not usable: it implies an attribute-only error of $\Phi(-6.5) \approx 4 \times 10^{-11}$, i.e. that the attributes classify the population essentially perfectly and the question of whether the network helps does not arise. A held-out linear discriminant on the same attributes attains a balanced error of 0.123. The discrepancy is the isotropy assumption: the per-dimension standard deviations span a factor of 28, so a single pooled σ is not meaningful, while $\|\Delta\|^2$ accumulates small differences over many correlated coordinates.

We therefore calibrate r from the achievable error rather than from the moments, taking $r = -2\Phi^{-1}(0.123) = 2.32$, and we use that value throughout Table 4. Resampling the split moves r by under 0.02, which changes no verdict in the table. This matters: at $r = 13$ the mixture term $\rho(1-\rho)\|\Delta\|^2$ reaches $1.67\sigma^2$ and the noise-dominated form (21) is invalid, whereas at $r = 2.32$ it is $0.05\sigma^2$ and the approximation is defensible. Any application of these thresholds should report r and how it was obtained.

On this population, every relation is one the network harms. The last column of Table 4 is below 1 for all eight relations: under the exact criterion, neighbour averaging on MGTAB is worse than using attributes alone on *every* recorded relation, by factors between 1.3 and 3.6. We report this as the principal empirical finding, and we note that an earlier version of this analysis reached the opposite conclusion for the social relations by applying the noise-dominated window (21) at $r = 13$; the "informative again with reversed sign" regime that produced is an artefact of using an $r \to 0$ approximation at large r, and it does not survive the exact criterion. Where the harmful interval does have an upper edge at $r = 2.32$—shared URL $(0.432, 0.995)$ and shared hashtag $(0.585, 0.868)$—the measured $\hat{\rho}$ falls inside it.

The layers still differ, in degree of harm and in mechanism. What survives, and is not an artefact, is the contrast between the two layers. On every social relation the automated accounts are almost perfectly camouflaged: $\tilde{\rho} = 1.00$, so the median automated account has *no* automated alter, and automated–automated ties number 212 in the following-and-friendship union against 40,263 automated–genuine ties. On the content relations the same accounts behave in the opposite way: $\hat{\rho}$ falls to 0.60–0.63 and automated–automated ties rise to 15,963 (shared URL) and 28,203 (shared hashtag). These accounts avoid one another where ties are visible and converge where content is shared. The ratio $\hat{\rho}_{\text{social}}/\hat{\rho}_{\text{content}}$ is 1.68, and the resulting harm differs correspondingly: the SNR ratio ranges from 0.79 on friendship to 0.44 on shared URLs.

What attribute-based tie scoring achieves. Section 4.4 bounds the tolerance attainable by a threshold rule on attribute discrepancy; we report the values here. On YelpChi homo ($c = 0.170$, $\tilde{D} = 158$, $r = 1.17$, $d = 32$) the optimum sits at $\bar{e}_w = 0.066$, $\bar{e}_c = 0.079$ and moves the tolerance from 0.7867 to 0.7875; on Amazon homo ($c = 0.074$, $\tilde{D} = 145$, $r = 2.04$, $d = 25$) from 0.8539 to 0.8634. Both remain below the measured $\hat{\rho}$ of 0.805 and 0.897, so on these networks attribute-discrepancy scoring does not restore the inference, whereas a flawless pruner would. The optimum is interior—it falls to zero as $\bar{e}_w \to 1$, because a threshold rule that discards nearly every tie leaves nothing to average—and is stable under enlarging the search domain, which the scale-invariant soft-weight model of (13) is not.

Multiplexity, not density, decides usability. These relations are recorded on the *same* 10,199 accounts, and both $\hat{\rho}$ and the crossover differ substantially between them. A procedure that pools relations into a single adjacency is therefore averaging quantities with different thresholds, which is the concrete form of the concern raised by Vörös and Snijders [42] and by Snijders et al. [43]: the quantity is a property of a relation, not of a population.

The degree-corrected identity is exact, once the estimators match. On the social union, the identity in its degree-corrected form $\eta = c\rho\,(\bar{d}_B/\bar{d}_H)$ predicts 0.0647508 against a measured 0.0647508: agreement to machine precision. This is not a coincidence but a consequence of what (2) is—an accounting identity over edge endpoints—and it holds on *any* graph provided ρ, η and the mean degrees are all estimated as ratios of sums, matching the sums the derivation counts. The uncorrected form $\eta = c\rho$ predicts 0.365, so the degree factor $\bar{d}_B/\bar{d}_H = 0.177$ is doing essential work here: automated accounts have mean degree 14.8 against 83.5 for genuine ones.

We flag the estimator sensitivity because it caught us out. Computing $\hat{\eta}$ as a *mean of per-node ratios*, while predicting it from a *ratio of means*, yields 0.032 against a predicted 0.067 and the appearance of a factor-of-two failure. The discrepancy is entirely an artefact of mixing the two estimators, compounded by averaging degrees over connected nodes only. Since a mean of ratios weights low-degree actors equally with hubs, the two estimators diverge exactly in proportion to within-class degree skew, which on this population is severe. The practical guidance is that (2) is a constraint an analyst can *check* on their own data, and a failed check indicates an estimator mismatch or a violated regularity assumption rather than a violated identity.

Caveat. The window is derived for a CSBM with regular degrees, isotropic Gaussian features, and a single relation. Real graphs violate all three: degrees are heavy-tailed, features are neither Gaussian nor isotropic, and the relations interact. The numbers above should be read as locating these graphs approximately, not as a certified verdict, and the agreement of η with $c\rho$ on YelpChi is the one quantity here that constitutes a genuine test of the theory rather than an application of it. Measuring ρ on TwiBot remains the natural next step and requires only data access; `theory/measure_rho.py` implements the estimator and validates it against planted ground truth, recovering ρ exactly and confirming $\eta = c\rho$ on synthetic graphs with regular degrees.

6. Discussion

6.1. From heuristic to criterion

The practical value of Corollary 8 is that it converts a design intuition into a quantity an operator can estimate. The inputs are the mean neighbourhood size D, the bot prevalence π_B, and the feature signal-to-noise ratio $\|\Delta\|/\sigma$—all measurable on a deployed system—and the output is a threshold on camouflage beyond which the graph is a liability. This reframes a question usually settled by ablation ("does the graph help on this benchmark?") as one that can be answered per-cohort: for hub accounts with large D the graph remains valuable at camouflage levels that make it actively harmful for sparse accounts. A detector that applies one aggregation policy to both is leaving accuracy on the table at one end and incurring damage at the other.

The analysis also isolates an under-appreciated coupling. Because $\eta = c\rho$, bot prevalence enters the critical ratio directly: a platform whose bot population doubles does not merely face more bots, it faces a detector whose tolerance to each bot's camouflage has fallen. Class imbalance is normally treated as a training-time nuisance to be fixed by resampling; (8) shows it is also a structural robustness parameter, and resampling does not touch it.

6.2. What the theory does and does not license

We are deliberate about the boundaries of these claims.

The results characterise *one round of mean aggregation* in a stylised generative model. They explain why homophilic message passing fails under camouflage and why the three mitigations help, and they give the correct qualitative orderings—more pruning is better, larger neighbourhoods are safer, higher prevalence is more dangerous. They do not predict the accuracy of a trained multi-layer network on a real bot graph, and no number in Section 5 should be read as such. Our simulations verify algebra; they are not empirical evidence about Twitter or any other platform.

Three specific gaps deserve naming. First, the single-relation restriction is real: our measurements show camouflage differs by a factor approaching two across relations of the same graph (Table 4), and a multi-relational treatment would need to model how a detector pools them. Second, although Section 4.5 extends the analysis to L layers, it does so in the noise-dominated regime and on a regular tree; a treatment that keeps the mixture terms at depth, and that handles the finite graphs where receptive fields saturate, is open. Third, Section 4.4 bounds the achievable pruning quality from the feature distributions but treats the scorer as a fixed threshold rule; bounding what a *learned* scorer attains, as a function of sample size and hypothesis class, is the natural next step and is what our real-data measurements suggest matters most and would turn Theorem 12 from a conditional into an unconditional guarantee.

6.3. Limitations of the generative model

The camouflage-CSBM makes four simplifications, each of which trades realism for tractability.

Isotropic, equal-covariance features. Real account features are neither isotropic nor identically dispersed across classes. Anisotropy would replace $\|\Delta\|/\sigma$ with a Mahalanobis distance throughout; the structure of the results survives, but $\rho^\star$ becomes direction-dependent.

Fixed neighbourhood size. Social graphs are heavy-tailed. Since $\rho^\star$ is concave in D, a mixture of degrees is *not* well summarised by its mean: applying (8) at the average degree overstates robustness for the low-degree majority. Per-degree stratification, which the theory directly supports, is the appropriate remedy.

Independent neighbour sampling. We ignore triadic closure and community structure, both prominent in real follow graphs. Correlated neighbourhoods reduce the effective sample size in the averaging step, which inflates the variance in (6) and lowers $\rho^\star$; our thresholds are therefore optimistic.

A single homogeneous relation. Production detectors operate on heterogeneous graphs with several edge types. The analysis extends per-relation, and the interesting question—how an adversary should allocate a fixed camouflage budget across relations—follows naturally from (8) but is beyond the present scope.

6.4. An adversarial reading

Finally, the theory can be read from the attacker's side, which we think is the more useful test of it. Corollary 8 tells an adversary exactly how much camouflage is needed to neutralise a homophilic detector, and Theorem 12 tells them how much more is needed against a pruning defence. That $\rho^\star_{\mathrm{SGTR}} \to 1$ only as $\bar{e}_c \to 1$ means no imperfect pruning rule is a permanent fix; the defence buys operating range, and an adversary willing to spend edges can always exceed it. Treating ρ as a strategic variable rather than a nuisance parameter—and analysing the resulting game between detector and adversary—strikes us as the most promising direction this line of work opens.

7. Conclusion

We close by restating the priority position plainly, because it determines how this paper should be cited. That a crossover exists—a camouflage level beyond which mean aggregation is worse than discarding the graph—is due to Ma et al. [5], and the fact that the harmful region is a bounded window rather than a half-line is the second root of their condition, discussed under the name mid-homophily pitfall by Luan et al. [15]. What we add is the class-imbalanced adversarial case: the edge-balance identity $\eta = c\rho$, which makes bot prevalence a robustness parameter and which we confirm on real data; a mixture-variance correction worth 11.5% of the threshold; the observation that pruning trades bias against effective degree, so that a bound on achievable edge-score quality is needed before any tolerance claim means anything; and the finding that mean aggregation's backtracking term acts as an implicit residual connection, so that the crossover rises with depth rather than being invariant to it.

We have given an analytical account of relation camouflage in GNN-based bot detection. Modelling camouflage as an adversarial parameter in a contextual stochastic block model, we proved that mean aggregation deflates the class signal by exactly $1-(1+c)\rho$, derived the resulting Bayes error, and obtained a critical camouflage ratio $\rho^\star = (1-D^{-1/2})/(1+c)$ beyond which a homophilic GNN is provably worse than ignoring the graph—a result that recovers the known $1/\sqrt{D}$ separability gain of graph convolution at $\rho = 0$. All closed forms were verified numerically, with the predicted critical ratio falling inside the measured crossover bracket.

The contribution is not a new detector but a criterion. Practitioners currently decide whether to trust a graph-based detector by running an ablation on a benchmark; the results here let them instead estimate three measurable quantities and compute the answer for their own deployment, per account cohort. Extending the analysis to multiple layers, to heterogeneous relations, and to a learned rather than assumed pruning quality would make that criterion sharper still.

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A. Deferred proofs

This appendix gives in full the proofs that are compressed or omitted in the main text. Throughout, π_B is the bot prevalence, $c = \pi_B/(1-\pi_B)$, $\eta = c\rho$, $\lambda = 1 - (1+c)\rho$, features are $\mathbf{x}_v \sim \mathcal{N}(\boldsymbol{\mu}_{y_v}, \sigma^2 I_d)$ with $\Delta = \boldsymbol{\mu}_B - \boldsymbol{\mu}_H$, and D is the neighbourhood size.

A.1. The edge-balance identity

Lemma 18. Let camouflage edges be those joining a bot to a human, and suppose each bot directs a fraction ρ of its D edges to humans. Then the expected fraction of a human's neighbourhood that is composed of bots is $\eta = c\rho$.

Proof. Let n be the number of nodes, so there are $\pi_B n$ bots and $(1-\pi_B)n$ humans. Each bot contributes ρD camouflage edge-endpoints, giving $\pi_B n\rho D$ endpoints in total. Every such endpoint lands on a human, and the humans collectively expose $(1-\pi_B)nD$ endpoints. Hence the expected bot fraction of a human neighbourhood is

$$\eta \;=\; \frac{\pi_B n\rho D}{(1-\pi_B)nD} \;=\; \frac{\pi_B}{1-\pi_B}\,\rho \;=\; c\rho.$$

The identity is a conservation statement about edge endpoints and is independent of how bots choose *which* humans to attach to; only the aggregate count matters. It presumes $\eta \le 1$, i.e. $\rho \le 1/c$, which is automatic when $\pi_B \le 1/2$. □

The asymmetry is the point: the adversary sets ρ directly but induces η on the other class, and the two coincide only when the classes are balanced ($c = 1$).

A.2. Proof of the depth result (Theorem 17)

Proof. Project onto $\hat{\mathbf{u}} = \Delta/\|\Delta\|$. Write $P = D^{-1}A$ for the simple-random-walk operator on the D-regular tree, so that L rounds of mean aggregation produce $\mathbf{h}^{(L)} = P^L\mathbf{x}$. Crucially P is *not* supported on the L-sphere: since $v \in \mathcal{N}(u)$ whenever $u \in \mathcal{N}(v)$, the walk backtracks, and $P^L e_v$ places mass at every distance $j \le L$ of the same parity as L, together with $j = 0$ when L is even.

Weights. By symmetry all nodes at distance j carry equal weight, so if $q_j^{(L)}$ denotes the probability that an L-step walk ends at distance j, each such node has weight $q_j^{(L)}/N_j$ with $N_j = D(D-1)^{j-1}$ and $N_0 = 1$. The distance process is the birth–death chain that steps outward with probability $(D-1)/D$ and inward with probability $1/D$ from any $j \ge 1$, and outward with certainty from $j = 0$; iterating it gives $q^{(L)}$.

Signal. Along a path in the tree the labels form a two-state Markov chain with transition matrix $\begin{pmatrix} 1-\rho & \rho \\ \eta & 1-\eta \end{pmatrix}$, whose eigenvalues are 1 and $\lambda = 1 - \rho -$

$\eta = 1-(1+c)\rho$. Hence for a node at distance j, $\mathbb{E}[s_w \mid y_v = B] - \mathbb{E}[s_w \mid y_v = H] = \lambda^j\|\Delta\|$, and summing against the weights gives $\Delta_L = \|\Delta\| \sum_j q_j^{(L)} \lambda^j$. Note this is *not* $\lambda^L\|\Delta\|$: the $j < L$ terms are deflated less, and the $j = 0$ term not at all.

Noise. For independent $\mathcal{N}(0, \sigma^2)$ noise, $\mathrm{Var}(\mathbf{h}_v^{(L)}) = \sigma^2\|P^L e_v\|^2 = \sigma^2 \sum_j N_j(q_j^{(L)}/N_j)^2 = \sigma^2 \sum_j (q_j^{(L)})^2/N_j$. Dividing signal by noise and normalising by $\mathrm{SNR}_{\mathrm{raw}} = \|\Delta\|/(2\sigma)$ gives (20); the absolute value is required because $\sum_j q_j \lambda^j$ can be negative. At $L = 1$, $q_1 = 1$ and $N_1 = D$, recovering $|\lambda|\sqrt{D}$. □

Remark 19 (Why the naive recursion fails, and by how much)**.** Propagating the deflation as $\Delta_{l+1} = \lambda\Delta_l$ and the variance as $V_{l+1} = V_l/D$ describes the *non-backtracking* walk, and yields $\mathrm{SNR}_L/\mathrm{SNR}_{\mathrm{raw}} = (\lambda\sqrt{D})^L$ and a depth-independent crossover. Both conclusions are artefacts. The true weight concentration $\sum_j (q_j^{(L)})^2/N_j$ exceeds D^{-L} by factors of 1.94, 4.63, 12.3 and 35.0 at $L = 2, 3, 4, 5$ for $D = 16$, and the crossover moves from 0.525 at $L = 1$ to 0.587 at $L = 2$. The direction is intuitive once stated: mass retained at distance $j < L$ is deflated only by λ^j, so backtracking raises the crossover. It does so at a cost, however: at $\rho = 0$ the same retained mass is over-weighted noise, and the non-backtracking walk attains a strictly higher SNR (60.0 against 29.7 at $L = 3$, $D = 16$). Backtracking trades separability in the homophilous regime for tolerance in the camouflaged one. A simulation that averages over the D^L leaves of a tree reproduces the naive recursion exactly and therefore cannot detect this error; one must simulate P^L.

A.3. Exact law of the edge score

Lemma 20. For an edge (u, v) let $s_{uv} = \|\mathbf{x}_u - \mathbf{x}_v\|^2$. Then $s \sim 2\sigma^2\chi_d^2$ if $y_u = y_v$, and $s \sim 2\sigma^2\chi_d^2(r^2/2)$ with $r = \|\Delta\|/\sigma$ otherwise. In particular $\mathbb{E}[s \mid \text{camouflage}] - \mathbb{E}[s \mid \text{within}] = \|\Delta\|^2$, and the ratio of the gap to the within-class standard deviation is $\kappa = r^2/(2\sqrt{2d})$.

Proof. Write $\mathbf{z} = \mathbf{x}_u - \mathbf{x}_v$. If $y_u = y_v$ then $\mathbf{z} \sim \mathcal{N}(0, 2\sigma^2 I_d)$, so $\|\mathbf{z}\|^2/(2\sigma^2) \sim \chi_d^2$. If $y_u \neq y_v$ then $\mathbf{z} \sim \mathcal{N}(\pm\Delta, 2\sigma^2 I_d)$ and $\|\mathbf{z}\|^2/(2\sigma^2)$ is noncentral chi-square with d degrees of freedom and noncentrality $\|\Delta\|^2/(2\sigma^2) = r^2/2$.

For the moments, $\mathbb{E}[\chi_d^2] = d$ and $\mathbb{E}[\chi_d^2(\nu)] = d + \nu$, so the means are $2\sigma^2 d$ and $2\sigma^2 d + \|\Delta\|^2$, giving the stated gap. Since $\mathrm{Var}(\chi_d^2) = 2d$, the within-class variance of s is $4\sigma^4 \cdot 2d$, whose square root is $2\sigma^2\sqrt{2d}$; dividing the gap by it gives $\kappa = \|\Delta\|^2/(2\sigma^2\sqrt{2d}) = r^2/(2\sqrt{2d})$.

The $d^{-1/2}$ dependence is the concentration phenomenon: the gap between the two score distributions grows like $\|\Delta\|^2$ while their spread grows like $\sqrt{d}$, so a fixed feature separation becomes progressively harder to detect from raw squared distances as the ambient dimension increases. □

B. Reproducibility

All closed forms in the paper are checked numerically by the scripts in `theory/`. `csbm_validation.py` and `refine_and_thca.py` cover Proposition 4 through Theorem 12; `corrections.py` covers Proposition 14, Proposition 15 and Theorem 17, and supersedes the earlier `extensions2.py`, which is retained only so that the two errors documented in Remarks 16 and A.2 can be reproduced; `measure_rho.py` implements the camouflage estimator of Section 5.5 together with a self-test against planted ground truth; `mgtab.py` produces Table 4 from the released MGTAB tensors, `exact_window.py` evaluates the exact criterion reported there, and `real_data.py` produces the review-network comparison. Each writes a JSON record of every figure quoted in the text. The scripts depend only on `numpy` and `scipy`; the fraud graphs are fetched from public URLs given in the script.